

*Non-Calculator***Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this quiz.****You have 30 minutes.**\_\_\_\_\_  
Signature**SHOW ALL WORK FOR FULL CREDIT!**

Find the product. Write the product in rectangular form, using exact values. (3 points)

1.  $\left[ 2\sqrt{3} \left( \cos \frac{13\pi}{12} + i \sin \frac{13\pi}{12} \right) \right] \left[ \sqrt{3} \left( \cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} \right) \right]$

Convert to trigonometric form, then divide. Write the quotient in rectangular form, using exact values. (3 points)

2.  $\frac{-2 - 2i\sqrt{3}}{\sqrt{2} - i\sqrt{6}}$

Find the power. Write the answer in rectangular form, using exact values. (3 points)

3.  $\left[ \sqrt{2} \left( \cos \frac{7\pi}{12} + i \sin \frac{7\pi}{12} \right) \right]^4$

Find all complex number solutions of the equation. Write answers in rectangular form. (3 points)

4.  $x^4 + 4 = 0$

The fractal called the **Mandelbrot set**: to determine if a complex number  $z = a + bi$  is in this set, repeatedly compute  $z, z^2 + z, (z^2 + z)^2 + z, [(z^2 + z)^2 + z]^2 + z, \dots$ . The complex number  $z$  does **not** belong to the Mandelbrot set if any of the resulting absolute values exceeds 2. Otherwise  $z$  is in the set. Determine whether or not each number belongs to the Mandelbrot set. (3 points)

|                |          |             |          |
|----------------|----------|-------------|----------|
| 5. a) $z = -i$ | YES / NO | b) $z = 2i$ | YES / NO |
|                |          |             |          |

## ANSWER KEY – Non-Calculator

## SHOW ALL WORK FOR FULL CREDIT!

Left: the work, one step per line, the way it goes on paper. Right: why that line happens.

Find the product. Write the product in rectangular form, using exact values. (3 points)

$$1. \left[ 2\sqrt{3} \left( \cos \frac{13\pi}{12} + i \sin \frac{13\pi}{12} \right) \right] \left[ \sqrt{3} \left( \cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} \right) \right]$$

## WORK

$$\begin{aligned} &= 2\sqrt{3} \cdot \sqrt{3} \operatorname{cis} \left( \frac{13\pi}{12} + \frac{5\pi}{4} \right) \\ &= 6 \operatorname{cis} \left( \frac{13\pi}{12} + \frac{15\pi}{12} \right) \\ &= 6 \operatorname{cis} \frac{28\pi}{12} = 6 \operatorname{cis} \frac{7\pi}{3} \\ &= 6 \operatorname{cis} \frac{\pi}{3} \\ &= 6 \left( \frac{1}{2} + \frac{\sqrt{3}}{2} i \right) \\ &= \boxed{3 + 3\sqrt{3}i} \end{aligned}$$

Trap:  $\frac{13\pi}{12}$  isn't on the unit circle, but the sum is. Add first, then look.

## WHY

Product theorem: multiply the  $r$ 's, add the  $\theta$ 's.

$$2\sqrt{3} \cdot \sqrt{3} = 2 \cdot 3 = 6. \quad \frac{5\pi}{4} = \frac{15\pi}{12}.$$

Simplify.

$$\frac{7\pi}{3} - \frac{6\pi}{3} = \frac{\pi}{3} \text{ (subtract } 2\pi\text{)}.$$

Unit circle at  $\frac{\pi}{3}$ .

Convert to trigonometric form, then divide. Write the quotient in rectangular form, using exact values. (3 points)

$$2. \frac{-2 - 2i\sqrt{3}}{\sqrt{2} - i\sqrt{6}}$$

## WORK

$$\begin{aligned} r_1 &= \sqrt{(-2)^2 + (-2\sqrt{3})^2} = \sqrt{4 + 12} = 4 \\ \theta_1 &= \tan^{-1} \left( \frac{-2\sqrt{3}}{-2} \right) = \tan^{-1}(\sqrt{3}) = 60^\circ \\ \theta_1 &= 60^\circ + 180^\circ = 240^\circ \\ r_2 &= \sqrt{(\sqrt{2})^2 + (-\sqrt{6})^2} = \sqrt{2 + 6} = 2\sqrt{2} \\ \theta_2 &= \tan^{-1} \left( \frac{-\sqrt{6}}{\sqrt{2}} \right) = \tan^{-1}(-\sqrt{3}) = -60^\circ \\ \theta_2 &= -60^\circ + 360^\circ = 300^\circ \\ \frac{4 \operatorname{cis} 240^\circ}{2\sqrt{2} \operatorname{cis} 300^\circ} &= \frac{4}{2\sqrt{2}} \operatorname{cis} (240^\circ - 300^\circ) \\ &= \frac{2}{\sqrt{2}} \operatorname{cis} (-60^\circ) \\ &= \sqrt{2} \operatorname{cis} 300^\circ \\ &= \sqrt{2} \left( \frac{1}{2} - \frac{\sqrt{3}}{2} i \right) \\ &= \boxed{\frac{\sqrt{2}}{2} - \frac{\sqrt{6}}{2} i} \end{aligned}$$

## WHY

Top first.

Trap: both negative is QIII, not  $60^\circ$ .

Bottom.

QIV: add  $360^\circ$ .

Quotient theorem: divide the  $r$ 's, subtract the  $\theta$ 's.

$$\frac{2}{\sqrt{2}} = \sqrt{2}. \quad -60^\circ + 360^\circ = 300^\circ.$$

QIV: cos +, sin -.

$$\sqrt{2} \cdot \sqrt{3} = \sqrt{6}.$$

Find the power. Write the answer in rectangular form, using exact values. (3 points)

3.  $\left[ \sqrt{2} \left( \cos \frac{7\pi}{12} + i \sin \frac{7\pi}{12} \right) \right]^4$

WORK

$$\begin{aligned} &= (\sqrt{2})^4 \operatorname{cis} \left( 4 \cdot \frac{7\pi}{12} \right) \\ &= 4 \operatorname{cis} \frac{28\pi}{12} = 4 \operatorname{cis} \frac{7\pi}{3} \\ &= 4 \operatorname{cis} \frac{\pi}{3} \\ &= 4 \left( \frac{1}{2} + \frac{\sqrt{3}}{2} i \right) \\ &= \boxed{2 + 2\sqrt{3}i} \end{aligned}$$

Trap:  $\frac{7\pi}{12}$  isn't a unit-circle angle, but  $4 \cdot \frac{7\pi}{12}$  reduces to one. Multiply first, then reduce.

WHY

De Moivre:  $r^n$  and  $n\theta$ .

$(\sqrt{2})^4 = 4$ . Simplify the angle.

$\frac{7\pi}{3} - \frac{6\pi}{3} = \frac{\pi}{3}$  (subtract  $2\pi$ ).

Unit circle at  $\frac{\pi}{3}$ .

Find all complex number solutions of the equation. Write answers in rectangular form. (3 points)

4.  $x^4 + 4 = 0$

WORK

$$x^4 = -4$$

$$r = 4, \quad \theta = 180^\circ$$

$$x = \sqrt[4]{4} \operatorname{cis} \left( \frac{180^\circ + 360^\circ k}{4} \right)$$

$$= \sqrt{2} \operatorname{cis} (45^\circ + 90^\circ k)$$

$$k = 0: \sqrt{2} \operatorname{cis} 45^\circ = \sqrt{2} \left( \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} i \right) = 1 + i$$

$$k = 1: \sqrt{2} \operatorname{cis} 135^\circ = \sqrt{2} \left( -\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} i \right) = -1 + i$$

$$k = 2: \sqrt{2} \operatorname{cis} 225^\circ = \sqrt{2} \left( -\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} i \right) = -1 - i$$

$$k = 3: \sqrt{2} \operatorname{cis} 315^\circ = \sqrt{2} \left( \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} i \right) = 1 - i$$

$$\boxed{x = 1 + i, -1 + i, -1 - i, 1 - i}$$

Trap:  $x = \pm\sqrt{2}i$  is wrong:  $(\sqrt{2}i)^4 = 4$ , not  $-4$ . The roots sit between the axes.

WHY

Isolate  $x^4$ .

$-4 + 0i$  sits on the negative real axis.

$n$ th root theorem,  $k = 0, 1, 2, 3$ .

$\sqrt[4]{4} = \sqrt{2}$ . Fourth roots sit  $90^\circ$  apart.

$\sqrt{2} \cdot \frac{\sqrt{2}}{2} = 1$ .

QII.

QIII.

QIV.

The fractal called the **Mandelbrot set**: to determine if a complex number  $z = a + bi$  is in this set, repeatedly compute  $z$ ,  $z^2 + z$ ,  $(z^2 + z)^2 + z$ ,  $[(z^2 + z)^2 + z]^2 + z, \dots$ . The complex number  $z$  does **not** belong to the Mandelbrot set if any of the resulting absolute values exceeds 2. Otherwise  $z$  is in the set. Determine whether or not each number belongs to the Mandelbrot set. (3 points)

5. a)  $z = -i$

YES / NO

WORK

a)  $z = -i: | -i | = 1$

$$z^2 + z = (-i)^2 - i = -1 - i$$

$$| -1 - i | = \sqrt{2} \approx 1.4$$

$$(-1 - i)^2 - i = (1 + 2i + i^2) - i = i$$

$$|i| = 1$$

$$i^2 - i = -1 - i \quad \boxed{\text{YES}}$$

b)  $z = 2i: |2i| = 2$

$$z^2 + z = (2i)^2 + 2i = -4 + 2i$$

$$| -4 + 2i | = \sqrt{16 + 4} = \sqrt{20} \approx 4.5 \quad \boxed{\text{NO}}$$

b)  $z = 2i$

YES / NO

WHY

Not over 2.

$$(-i)^2 = i^2 = -1.$$

Not over 2, keep going.

Square the last result, add  $z$ .

It repeats  $(-1 - i, i, \dots)$  and never passes 2.

Exactly 2 is not over 2: keep going.

$$(2i)^2 = 4i^2 = -4.$$

Over 2: stop.